RC & RA

CPE241 Database System

Example Domain: Medical

Patient(pID, pName, pAge)
Doctor(dID, dName, dAge)
Appointment(pID, dID, aDatetime)

Basic Queries

Select patients with name = 'Adam'

RA

$\sigma_{pName='Adam'}(Patient)$

TRC

$\{p \mid p \in Patient \land p.pName='Adam'\}$

DRC

$\{pID, pName, pAge \mid Patient(pID, pName, pAge) \land pName='Adam'\}$

SQL

```
SELECT *
FROM Patient
WHERE pName = 'Adam';
```

Add another condition

“Find patient whose name is ‘Adam’ and age > 50”

RA

$\sigma_{pName='Adam' \land pAge>50}(Patient)$

TRC

$\{p \mid p \in Patient \land p.pName='Adam' \land p.pAge>50\}$

DRC

$\{pID, pName, pAge \mid Patient(pID, pName, pAge) \land pName='Adam' \land pAge>50\}$

SQL

```
SELECT * FROM Patient
WHERE pName = 'Adam'
  AND pAge > 50;
```

Projection

Find pID of patients whose name is ‘Adam’ and age > 50

RA

$\pi_{pID}(\sigma_{pName='Adam' \wedge pAge>50}(Patient))$

TRC

$\{p.pID \mid p \in Patient \wedge p.pName='Adam' \wedge p.pAge>50\}$

DRC

$\{pID \mid \exists\, pName, pAge(Patient(pID, pName, pAge) \wedge pName='Adam' \wedge pAge>50)\}$

SQL

```
SELECT pID FROM Patient
WHERE pName = 'Adam'
    AND pAge > 50;
```

Join them up

Find the name of patients who have an appointment today

RA

$\pi_{pName}(\sigma_{date(aDatetime)=today()}(Patient \bowtie Appointment))$

TRC

$\{p.pName \mid p \in Patient \wedge a \in Appointment \wedge p.pID=a.pID \wedge date(a.aDatetime)=today()\}$

DRC

$\{pName \mid \exists pID,pAge,dID,aDatetime(Patient(pID,pName,pAge) \wedge Appointment(pID,dID,aDatetime) \wedge date(aDatetime)=today())\}$

SQL

```
SELECT DISTINCT p.pName FROM Patient p
JOIN Appointment a ON p.pID = a.pID
WHERE DATE(a.aDatetime) = CURRENT_DATE;
```

Not Existing

Find name of patients who have no appointment at all

RA

$\pi_{pName}(Patient) - \pi_{pName}(Patient \bowtie Appointment)$

TRC

$\{p.pName \mid p \in Patient \land \neg \exists a \in Appointment(a.pID = p.pID)\}$

DRC

$\{pName \mid \exists pID, pAge(Patient(pID, pName, pAge) \land \neg \exists dID, aDatetime(Appointment(pID, dID, aDatetime)))\}$

SQL

```
SELECT p.pNameFROM Patient p
WHERE NOT EXISTS
(SELECT * FROM Appointment a WHERE a.pID = p.pID)
```

All

Find name of patients who have an appointment with every doctor

RA

$\pi_{pName}(Patient \bowtie (Appointment(pID,dID) \div Doctor(dID)))$

TRC

$\{p.pName \mid p \in Patient \wedge \neg \exists d \in Doctor(\neg \exists a \in Appointment(a.pID=p.pID \wedge a.dID=d.dID))\}$

DRC

$\{p.pName \mid p \in Patient \wedge \forall d \in Doctor \exists a \in Appointment(a.pID=p.pID \wedge a.dID=d.dID)\}$

SQL

```
SELECT p.pName FROM Patient p
WHERE NOT EXISTS (
    SELECT * FROM Doctor d
    WHERE NOT EXISTS (
    SELECT * FROM Appointment a
    WHERE a.pID = p.pID
    AND a.dID = d.dID
    )
);
```

Schema 1: Employee / Manager (Self-relationship)

Employee(EID, EName, Dept, Salary, ManagerID)

Practice queries

1. Employees with **no manager** ✅
2. Employees who **manage no one** ✅
3. Employees whose **manager works in the same department**
4. Highest-paid employee(s) ❗ ($\neg\exists$ or $\forall$)
5. Employees who earn **more than their manager** ❗
6. (Challenge) Managers who manage **all employees in their department**

Schema 2: Student / Course / Registration

Student(SID, SName, Major, Year)
Course(CID, CName, Credits, Dept)
Register(SID, CID, Grade)

Practice queries

1. Students who register **at least one course** ✅
2. Students who register **no course** ✅
3. Students who register **"Database Systems"** ✅
4. Students who get **only A grades** ❗
5. Students who register **ALL courses in a department** ✅✅ (division)
6. (Challenge) Courses that **all students** register for ❗❗

Schema 3: Product / Order / Payment

Product(PID, PName, Price)
Order(OID, OrderDate)
OrderItem(OID, PID, Quantity)
Payment(OID, PayDate, Amount)

Practice queries

1. Products **never ordered** ✅
2. Orders with **no payment** ✅
3. Products ordered **today**
4. Customers/orders with **multiple products**
5. Orders where **total payment < total order cost** ❗
6. (Challenge) Products ordered in **every order** ✅✅ (division)